

ENUNCIADO — PROBLEMA 1

Department of Mathematical Sciences
George Mason University

Jan 12, 2011

Numerical Analysis - Preliminary Exam.

Please do FIVE of the following seven problems.
Closed books, closed notes. Show work for full credit. No calculators allowed.

Problem 1 (a) Use the Newton-Raphson method: $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$ to establish the following iterative formula for finding $\sqrt{9}$: $x_{n+1} = \frac{x_n^2 + 9}{2x_n}, n > 0$.

(b) Show that if the sequence x_n converges to $x^* = 3$, the convergence rate is quadratic.

Problem 2 Provide a rigorous proof of the fact that if Gaussian elimination with partial pivoting is applied to a matrix A that is diagonally dominant by columns, then no row interchanges will occur.

Problem 3 (a) Consider the formula $\int_0^h f(x) dx = h \left\{ Af(0) + Bf\left(\frac{h}{3}\right) + Cf(h) \right\}$. Find A, B, C such that this formula is exact for all polynomials of degree less than or equal to 2.

(b) Suppose that the Trapezoidal rule $\int_0^h f(x) dx = \frac{h}{2} \{f(0) + f(h)\}$ applied to $\int_0^2 f(x) dx$ gives the value $\frac{1}{2}$ while the quadrature rule in part (a) applied to the same integral gives the value $\frac{1}{4}$. If $f(0) = 3$, then show that this implies that $f\left(\frac{2}{3}\right) = 1$.

Problem 4 Use the linear ODE $x' = \lambda x$ to analyze accuracy and stability of the Heun's method $x_{k+1} = x_k + \frac{h_k}{2}(k_1 + k_2)$, where $k_1 = f(t_k, x_k), k_2 = f(t_k + h_k, y_k + h_k k_1)$. Prove that the method is 2nd order accurate and characterize its stability region.

Problem 5 Suppose that the altitude of the trajectory of a projectile is described by the 2nd order ODE $u'' = -4$. Suppose that the projectile is fired from position $t = 0$ and height $u(0) = 1$ and is to strike a target at position $t = 1$, also of height $u(1) = 1$.

(a) Solve this BVP by the shooting method: use the trapezoid rule with step $h = 1$ to derive a system of 2 equations to determine the initial slope at $t = 0$ required to hit the desired target at $t = 1$. Using the initial slope found above and a step size $h = 0.5$, estimate the projectile height at $t = 0.5$.

(b) Solve the same BVP using a finite difference method with $h = 0.5$. Compare the height estimation at $t = 0.5$ with that obtained via shooting method.

SOLUCIÓN — PROBLEMA 1

Problema 1 — Método de Newton (orden de convergencia)

(a) Para $\sqrt{9}$, tomamos $f(x) = x^2 - 9$, $f'(x) = 2x$. Newton-Raphson:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^2 - 9}{2x_n} = \frac{2x_n^2 - (x_n^2 - 9)}{2x_n} = \frac{x_n^2 + 9}{2x_n}.$$

(b) Sea $e_n = x_n - 3$. Entonces

$$x_{n+1} - 3 = \frac{x_n^2 + 9}{2x_n} - 3 = \frac{x_n^2 - 6x_n + 9}{2x_n} = \frac{(x_n - 3)^2}{2x_n} = \frac{e_n^2}{2x_n}.$$

Como $x_n \rightarrow 3$, se tiene $e_{n+1} \approx \frac{1}{6}e_n^2$, es decir $e_{n+1} = O(e_n^2)$: **convergencia cuadrática**, con constante asintótica $C = f''(3)/(2f'(3)) = 2/12 = 1/6$.

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Problema 2 — Eliminación gaussiana con pivoteo parcial y dominancia diagonal por columnas

Afirmación: si A es estrictamente diagonalmente dominante por columnas ($|a_{jj}| > \sum_{i \neq j} |a_{ij}|$ para cada columna j), entonces la eliminación gaussiana con pivoteo parcial nunca intercambia filas.

Demostración (inducción sobre el paso de eliminación).

Paso 1. Dominancia en la columna 1 da $|a_{11}| > \sum_{i \neq 1} |a_{i1}| \geq |a_{i1}|$ para todo $i \neq 1$ (suma de términos no negativos domina a cualquier sumando). Luego a_{11} ya es el elemento de mayor valor absoluto de su columna: el pivoteo parcial no intercambia filas en este paso.

Paso inductivo. Sea $S_1 = \sum_{i \geq 2} |a_{i1}| < |a_{11}|$. Los multiplicadores son $m_{i1} = a_{i1}/a_{11}$, y la submatriz actualizada es $a_{ij}^{(2)} = a_{ij} - m_{i1}a_{1j}$ para $i, j \geq 2$. Para cualquier columna $j \geq 2$:

$$|a_{jj}^{(2)}| \geq |a_{jj}| - \frac{|a_{j1}||a_{1j}|}{|a_{11}|}, \quad \sum_{i \geq 2, i \neq j} |a_{ij}^{(2)}| \leq \sum_{i \geq 2, i \neq j} |a_{ij}| + \frac{|a_{1j}|}{|a_{11}|} \sum_{i \geq 2, i \neq j} |a_{i1}|.$$

Sumando $|a_{j1}|$ a ambos lados de la segunda desigualdad se obtiene $\sum_{i \geq 2, i \neq j} |a_{i1}| + |a_{j1}| = S_1$, así que el término de corrección total es $(|a_{1j}|/|a_{11}|) S_1 < |a_{1j}|$ (porque $S_1 < |a_{11}|$).

Por dominancia original en columna j : $|a_{jj}| - \sum_{i \geq 2, i \neq j} |a_{ij}| > |a_{1j}|$ (ya que $|a_{jj}| > \sum_{i \neq j} |a_{ij}| = |a_{1j}| + \sum_{i \geq 2, i \neq j} |a_{ij}|$).

Combinando ambas desigualdades:

$$|a_{jj}^{(2)}| > |a_{jj}| - \sum_{i \geq 2, i \neq j} |a_{ij}| - \frac{|a_{1j}|}{|a_{11}|} S_1 > |a_{1j}| - |a_{1j}| = 0 \text{ (con margen estricto)} \Rightarrow |a_{jj}^{(2)}| > \sum_{i \geq 2, i \neq j} |a_{ij}^{(2)}|.$$

Es decir, la submatriz $(n-1) \times (n-1)$ restante vuelve a ser diagonalmente dominante por columnas. Por inducción, en **cada** paso el pivote natural ya es el máximo de su columna, así que el pivoteo parcial nunca reordena filas. ■

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Problema 3 — Cuadratura de 3 puntos

(a) Se buscan A, B, C tales que $\int_0^h f \, dx = h\{Af(0) + Bf(h/3) + Cf(h)\}$ sea exacta para $f = 1, x, x^2$:

- $f = 1$: $h = h(A + B + C) \Rightarrow A + B + C = 1$.
- $f = x$: $h^2/2 = h^2(B/3 + C) \Rightarrow B/3 + C = 1/2$.
- $f = x^2$: $h^3/3 = h^3(B/9 + C) \Rightarrow B/9 + C = 1/3$.

Restando: $(B/3 + C) - (B/9 + C) = 1/2 - 1/3 \Rightarrow \frac{2B}{9} = \frac{1}{6} \Rightarrow B = \frac{3}{4}$. Luego $C = \frac{1}{2} - \frac{B}{3} = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$, y $A = 1 - B - C = 0$.

$$\boxed{A = 0, \quad B = \frac{3}{4}, \quad C = \frac{1}{4}.}$$

(b) En $[0, 2]$: trapecio da $f(0) + f(2) = \frac{1}{2} \cdot 2 \dots$; con la fórmula $\frac{h}{2}\{f(0) + f(h)\}$, $h = 2$: $f(0) + f(2) = \frac{1}{2} \Rightarrow f(2) = \frac{1}{2} - 3 = -\frac{5}{2}$ (usando $f(0) = 3$).

Con la regla del inciso (a), $h = 2$: $\frac{2}{3}f(2/3) + \frac{1}{3}f(2) = \frac{1}{3} \Rightarrow f(2/3) = \frac{1}{2} - \frac{1}{3}f(2) = \frac{1}{2} - \frac{1}{3}(-\frac{5}{2}) = \frac{1}{2} + \frac{5}{6} = \frac{4}{3} \Rightarrow f(2/3) = \frac{4}{3}$. $\square$

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Problema 4 — Exactitud y estabilidad del método de Heun

Aplicado a $x' = \lambda x$: $k_1 = \lambda x_k$, $k_2 = \lambda(x_k + h_k \lambda x_k) = \lambda x_k(1 + h_k \lambda)$.

$$x_{k+1} = x_k + \frac{h_k}{2}(k_1 + k_2) = x_k \left[1 + h_k \lambda + \frac{1}{2}(h_k \lambda)^2 \right].$$

Con $z = h_k \lambda$, el factor de amplificación es $R(z) = 1 + z + \frac{1}{2}z^2$, que coincide con la expansión de Taylor de e^z hasta el término $z^2/2$ (el error es $O(z^3)$): esto confirma que el método es **de orden 2** (error local $O(h^3)$, global $O(h^2)$).

Región de estabilidad (eje real, $\lambda < 0$): resolver $|R(z)| \leq 1$. $R(z) = 1 \iff z(1 + z/2) = 0 \iff z = 0$ o $z = -2$. Como R tiene mínimo en $z = -1$ con $R(-1) = 0.5 > -1$, la cota inferior $R = -1$ nunca se alcanza en el eje real. Luego la región de estabilidad real es

$$-2 < h_k \lambda < 0,$$

es decir $h_k < 2/|\lambda|$ para λ real negativo.

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Problem 6 Consider solving a linear system $Ax = b$ using an iterative method

$$\mathbf{x}^{(k+1)} = B\mathbf{x}^{(k)} + \mathbf{c}$$

for $k \geq 0$ with a given initial approximation $\mathbf{x}^{(0)}$.

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Problem 7 (a) State the definition of an orthogonal matrix

(b) Find an orthogonal matrix Q and a number α such that

$$Q \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} \alpha \\ 0 \end{bmatrix}$$

SOLUCIÓN — PROBLEMA 5

Problema 5 — BVP: disparo (shooting) vs. diferencias finitas

$u'' = -4$, $u(0) = 1$, $u(1) = 1$. Solución exacta: $u(t) = -2t^2 + 2t + 1$ (con $u'(0) = 2$), y $u(0.5) = 1.5$.

(a) Shooting con regla del trapecio, $h = 1$. Sistema de primer orden: $u' = v$, $v' = -4$, con $u(0) = 1$, $v(0) = s$ desconocida. Trapecio (1 paso, $h = 1$):

$$v_1 = s + \frac{1}{2}(-4 - 4) = s - 4, \quad u_1 = 1 + \frac{1}{2}(v_0 + v_1) = 1 + \frac{1}{2}(2s - 4) = s - 1.$$

Imponiendo $u_1 = u(1) = 1$: $s - 1 = 1 \Rightarrow s = 2$ (pendiente inicial).

Con $s = 2$ y $h = 0.5$ (un paso de trapecio hasta $t = 0.5$):

$$v_1 = 2 + \frac{0.5}{2}(-4 - 4) = 2 - 2 = 0, \quad u_1 = 1 + \frac{0.5}{2}(2 + 0) = 1 + 0.5 = 1.5.$$

Estimación: $u(0.5) \approx 1.5$ (exacta aquí, porque el trapecio integra exactamente funciones lineales/cuadráticas y $v' = -4$ es constante).

(b) Diferencias finitas, $h = 0.5$. Nodos $t_0 = 0, t_1 = 0.5, t_2 = 1$, $u_0 = 1, u_2 = 1$. Diferencia central: $\frac{u_0 - 2u_1 + u_2}{h^2} = -4 \Rightarrow \frac{1 - 2u_1 + 1}{0.25} = -4 \Rightarrow u_1 = 1.5$.

Comparación: ambos métodos dan $u(0.5) = 1.5$, coincidiendo con el valor exacto — esperado porque el problema es cuadrático y ambos esquemas son exactos para polinomios de grado ≤ 2 (trapecio) o ≤ 3 (diferencia central de segundo orden) en este caso particular.

ENUNCIADO — PROBLEMA 6

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Problema 6 — Convergencia de métodos iterativos lineales

(a) Sea $e^{(k)} = x^{(k)} - x^*$. De $x^{(k+1)} = Bx^{(k)} + c$ y $x^* = Bx^* + c$ se sigue $e^{(k+1)} = Be^{(k)} = B^{k+1}e^{(0)}$. Si el método converge para **todo** $x^{(0)}$ (equivalentemente, para todo $e^{(0)}$), entonces $B^k \rightarrow 0$ como operador. Si $\rho(B) \geq 1$, existe λ con $|\lambda| \geq 1$ y vector propio $v \neq 0$; tomando $e^{(0)} = v$ da $e^{(k)} = \lambda^k v$, que no tiende a 0. Contradicción. Luego $\rho(B) < 1$ es necesario. ■

(b) De $x^{(k+1)} - x^{(k)} = B(x^{(k)} - x^{(k-1)})$ se sigue por inducción $\|x^{(k+n)} - x^{(k+n-1)}\| \leq \|B\|^n \|x^{(k)} - x^{(k-1)}\|$. Usando el telescopio $x^* - x^{(k)} = \sum_{n=1}^{\infty} (x^{(k+n)} - x^{(k+n-1)})$ y la desigualdad triangular:

$$\|x^{(k)} - x^*\| \leq \sum_{n=1}^{\infty} \|B\|^n \|x^{(k)} - x^{(k-1)}\| = \frac{\|B\|}{1 - \|B\|} \|x^{(k)} - x^{(k-1)}\|,$$

usando la serie geométrica (válida porque $\|B\| < 1$). ■

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Problema 7 — Matriz ortogonal

(a) Q es ortogonal si $Q^T Q = Q Q^T = I$ (equivalentemente $Q^{-1} = Q^T$); sus columnas (y filas) son vectores ortonormales.

(b) Como Q preserva la norma, $|\alpha| = \|(1, 2)^T\| = \sqrt{5}$. Con una rotación de Givens $Q = \begin{pmatrix} c & s \\ -s & c \end{pmatrix}$, $c = 1/\sqrt{5}$, $s = 2/\sqrt{5}$:

$$Q \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} c + 2s \\ -s + 2c \end{pmatrix} = \begin{pmatrix} 1/\sqrt{5} + 4/\sqrt{5} \\ -2/\sqrt{5} + 2/\sqrt{5} \end{pmatrix} = \begin{pmatrix} \sqrt{5} \\ 0 \end{pmatrix}.$$

$$\alpha = \sqrt{5}, \quad Q = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix}.$$